

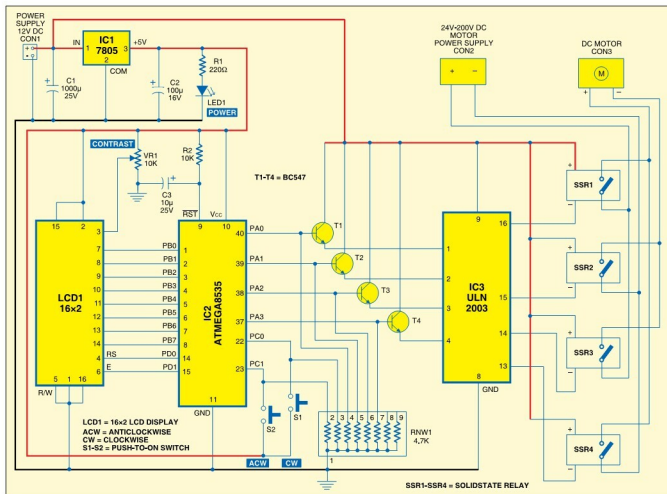


Fig. 3: Circuit of AVR-based high-current DC motor controller

interfaced with PD0 and PD1 pins of the microcontroller, respectively.

Port pins PA0 through PA3 of IC2 are connected to transistors T1



Fig. 4: Connections of the four SSRs

through T4, respectively. The emitters of these transistors are connected to the ULN2003 (IC3) to drive the solidstate relays. ULN2003 has seven drivers capable of sinking 500mA current and provides ground path to the solidstate relays.

The solidstate relays (SSR1 through SSR4) are used in H-Bridge configuration for controlling the direction of the DC motor. The controller energises SSR1 and SSR2 simultaneously for clockwise rotation, while SSR3 and SSR4 are energised for anticlockwise rotation.

The relays are used here to drive a 200V DC, 40A load. These can operate a wide range of DC motors in industrial applications. Each SSR has four terminals: two for 3-32V DC excitation and the other two for 24-200V

DC load with a current output of 40A. The connections of the four SSRs are shown in Fig. 4.

Input switches S1 and S2 are used to select the direction of rotation of the motor. The controller scans the logic-high state from the input key operation. When switch S1 is pressed, the corresponding pin is pulled high, 'clockwise' message is displayed on the LCD and the motor rotates in the clockwise direction. On pressing switch S2, 'anticlockwise' message is displayed on the LCD and the motor rotates in anticlockwise direction.

Software program

The microcontroller program is written in 'C' language using Image Craft ICC AVR software. The flow-chart of